STATS 507 Project Proposal: Spatiotemporal Forecasting of NYC Taxi Demand Using Deep Sequential Models

Brynn Woolley
PhD Student (NGTS)
Department of Civil Engineering
University of Michigan Ann Arbor, Ann Arbor
bwoolley@umich.edu

1 Overview

1.1 Background

Accurate short-term demand prediction is vital for optimizing urban transportation systems, including demand forecasting, congestion mitigation, and taxi dispatching. Traditional methods (e.g., ARIMA, regression analysis) often struggle to model complex temporal dependencies and spatial dynamics present in real-world mobility data. Deep learning models such as Long Short-Term Memory (LSTM) and Transformers offer better capacity to model these patterns, especially as transportation systems increasingly rely on emerging data streams, such as passively collected location-based services (LBS) from mobile phones.

New York City (NYC) generates millions of taxi trips every day. Each of these trips is equipped with detailed spatial and temporal metadata. This rich data can provide a better foundation for accurately predicting the demand on the transportation network, which, in turn, supports effective urban mobility planning and improves real-time dispatch optimization. In recent years, deep learning models (e.g., long short-term memory (LSTM) networks, transformers) have been found to be powerful tools for optimizing time-series forecasting tasks, making them particularly effective for analyzing and predicting taxi demand in the city.

While the NYC Yellow Taxi Trip dataset spans several years, preliminary analysis indicates that potentially only three days of data provide rich, detailed information. This limitation may restrict the temporal scope of the analysis and could impact the generalization of findings across broader time frames.

1.2 Motivation

Currently, my research focuses on understanding and addressing the bias and validation challenges associated with LBS data. My goal is to exploit the inherent strengths of LBS data (large sample sizes and longitudinal patterns)—while systematically addressing their known limitations (sparsity and inconsistent sampling rates). I am particularly interested in exploring how various modeling techniques can harness these strengths to yield more accurate demand predictions. In my exploration of the mechanics of these models, I hope to uncover how they can interpret the complex patterns present in transportation data. Understanding the application would deepen my insight into their predictive capabilities and allow me to determine real applications that facilitate urban mobility.

This project will use historical pickup data to predict hourly demand in different areas of New York City. We will analyze the NYC Yellow Taxi Trip dataset from Hugging Face. Using PyTorch, we aim to apply deep learning to uncover nuanced insights using advanced deep sequential models, such as LSTM and Transformer architectures. We aim to create a predictive model that can handle the changing patterns of demand and generalize across different spatial zones and time windows. This method will lead to better demand forecasting, helping to improve decision-making in city transportation planning.

Understanding how taxi demand changes over time is important for grasping New York City's transportation needs.This analysis will help us evaluate the effectiveness of advanced methods like LSTM networks and Transformers for short-term mobility predictions. Additionally, this examination will identify areas in the city with high demand fluctuations or steady demand patterns, which will aid in strategic planning and resource allocation.

1.3 Anticipated Insights

1. Temporal Pattern Characterization: Understand short-term demand variability within the three-day window.

2. Model Performance Differentiation: Evaluate LSTM vs. Transformer models.

3. Spatial Demand Dynamics: Identify zones with stable vs. volatile demand.

4. LBS Data Validation Insights: Reflect on how inconsistent sampling affects modeling and inference.

2 Prior Work

2.1 Literature Review

LSTM networks have been widely used for sequential prediction tasks due to their ability to capture long-term dependencies in time series. Abideen et al. (2021) applied LSTM variants to model taxi driver behavior and next-destination prediction. Cao et al. (2022) compared LSTM with Transformer-based approaches, highlighting LSTM's strength in short-term predictions. LSTM often serves as a benchmark for more recent Transformer-based methods in demand forecasting. Initially designed for NLP, transformers have been adapted for time series via self-attention mechanisms that efficiently model long-range dependencies.

Initially designed for NLP, transformers have been adapted for time series via self-attention mechanisms that efficiently model long-range dependencies. Bi et al. (2022) used a Transformer network for online car-hailing demand prediction, outperforming LSTM and ConvLSTM baselines. Cheng et al. (2023) combined Swin Transformer and LSTM for multivariate urban mobility forecasting on NYC and TaxiBJ datasets. Transformers often scale better for city-wide, multivariate demand modeling than LSTM-based models.

Recent architectures aim to capture both spatial and temporal dependencies by combining CNN, LSTM, and Transformer components. Ling et al. (2023) introduced STHAN (Spatiotemporal Meta Graph Transformer), which uses compound attention layers for fine-grained spatiotemporal learning. Wen et al. (2023) proposed the Sandwich Transformer, integrating Bi-LSTM for learning periodic signals and attention-based encoders for modeling city-wide demand shifts. These hybrid models reflect an ongoing trend toward fusing domain-specific priors (e.g., spatiotemporal structure) with general-purpose sequence architectures for improved forecasting accuracy.

2.2 Potential Methods

To predict short-term taxi demand, this project will evaluate and compare the performance of LSTM and Transformer-based deep learning models. Both models will be trained using historical NYC taxi pickup data aggregated by hour and spatial grid cell. The LSTM model will serve as a strong baseline due to its effectiveness in short-horizon time—series tasks, while the Transformer model-adapted for time series using positional encodings or time2vec embeddings—will be assessed for its ability to capture longer-range dependencies and more complex temporal dynamics.

Datasets such as NYC Yellow Taxi, TaxiBJ, and CDtaxi are widely used for zone-level mobility modeling, and this project adopts a similar structure by framing the prediction task at the spatial grid and hourly resolution. Most prior studies evaluate model performance using metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), and visual techniques such as heatmaps and time-series plots to illustrate demand variability and model fit. This project will follow that evaluation approach.

Spatial encoding will initially rely on a fixed grid mapping of latitude and longitude coordinates. Depending on early results, the project may explore spatial attention mechanisms or lightweight convolutional layers to incorporate local spatial context and neighborhood-level interactions.

The comparison will emphasize generalization performance under limited temporal data and the models' capacity to capture short-term spatiotemporal demand patterns across NYC zones.

3 Preliminary Results

3.1 Data Description

We use the NYC Yellow Taxi Trip dataset hosted on Hugging Face. The data includes timestamps, pickup and dropoff locations, and trip metadata. For this project, we focus on aggregating hourly demand at the zone level using the `PULocationID` field.

Initial inspection shows regular weekly and daily periodicity in total pickups, with significant variance across spatial zones. Figure 1 shows the total hourly pickups citywide on a log scale, highlighting strong diurnal cycles. Figure 2 presents hourly demand trends for the top 5 most active pickup zones.

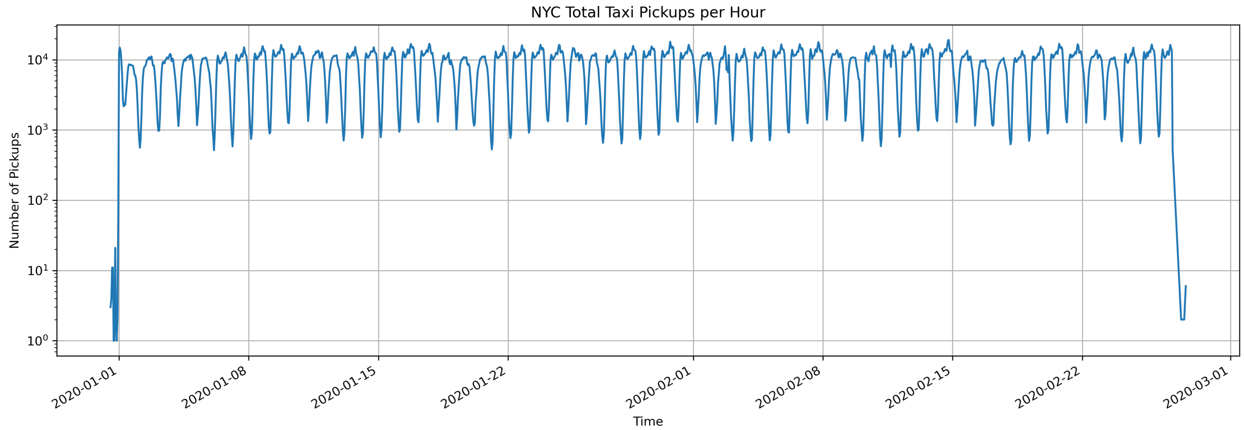

Figure 1: Hourly total NYC taxi pickups from Jan–Feb 2020 (log scale).

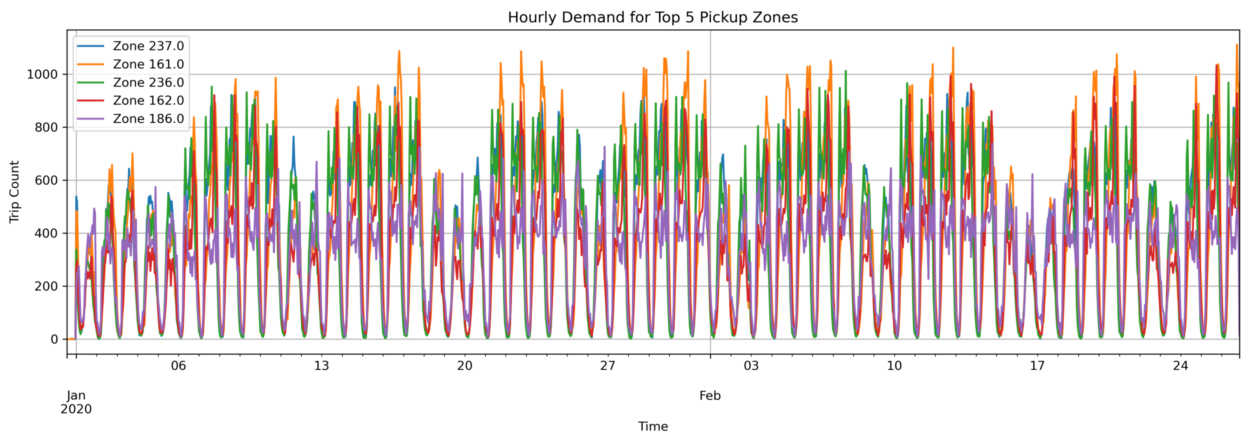

Figure 2: Hourly demand trends for the top 5 pickup zones.

Data Preprocessing:

- Aggregated trip counts by hour and zone (`PULocationID`).
- Engineered temporal features: hour-of-day, day-of-week.
- Added lagged demand features for autoregressive modeling.
- Normalized inputs for model training.

Observed Challenges:

- Sparse or zero demand in low-activity zones introduces variance.
- Demand trends shift across weekdays and weekends.
- Need to balance model complexity with short training window.

3.2 Model Description:

- **LSTM**: a univariate sequence model trained on historical hourly demand per zone.
- **Transformer Encoder**: adapted with fixed positional encoding and attention over hourly input sequences.

Training Setup:

- Target: next-hour demand per zone.
- Features: lagged demand, hour-of-day, day-of-week.
- Loss Function: Mean Squared Error (MSE).
- Evaluation Metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and visual inspection via time-series plots.

Future work includes:

- Hyperparameter tuning and model regularization.
- Multi-zone modeling with spatiotemporal attention.
- Potential incorporation of exogenous features (weather, holidays).

4 Potential Tools:

This project utilizes a combination of foundational data science tools and deep learning frameworks introduced throughout the course, along with additional libraries for advanced modeling.

Data Handling and Preprocessing:

- `pandas`: For reading, filtering, grouping, and aggregating structured data (e.g., trip-level to hourly zone-level demand).
- `numpy`: For numerical operations and array manipulations during feature engineering.
- `datetime`: For timestamp parsing and temporal feature extraction.

Exploratory Analysis and Visualization:

- `matplotlib`, `seaborn`: For visualizing hourly demand trends, spatial variability, and residual plots.
- Time-series plots and heatmaps are used to assess temporal dynamics and model performance.

Modeling and Evaluation:

- `PyTorch`: For implementing deep learning models, including LSTM and Transformer-based architectures.
- `scikit-learn`: For preprocessing (e.g., feature scaling), splitting datasets, and computing performance metrics (e.g., MAE, RMSE).

- `Hugging Face Transformers (optional)`: May be explored for pre-built transformer components adapted to time series tasks.

Additional Tools (Planned):

- `PyTorch Lightning`: To streamline model training, checkpointing, and experimentation.
- `Optuna` or `Ray Tune` (optional): For hyperparameter optimization.
- `geopandas` (optional): For spatial mapping or integration with NYC Taxi Zone shapefiles.

These tools provide a complete pipeline from raw mobility data to trained demand forecasting models, and they enable scalable experimentation with different architectures and feature sets.

5 Project Deliverables

The final deliverables will consist of a complete implementation pipeline, evaluation outputs, documentation, and a concise summary report. These deliverables are designed to meet course expectations while also aligning with best practices in applied mobility modeling.

- **GitHub Repository:**
  - Well-documented and reproducible codebase using Python.
  - Jupyter notebooks and/or Python scripts for data preprocessing, model training, evaluation, and visualization.
  - Requirements file (`requirements.txt`) for setting up dependencies.
- **Final Models:**
  - Trained LSTM and Transformer models for short-term taxi demand forecasting.
  - Evaluation artifacts (e.g., prediction outputs, saved weights, training logs).
  - Visualizations comparing model predictions vs. ground truth.
- **Evaluation Results:**
  - Quantitative results using RMSE, MAE, and potentially MAPE.
  - Qualitative visualizations (e.g., time-series plots, error trends, zone-level demand heatmaps).
- **Two-Page Report:**
  - A concise summary (`summary.md` or PDF) that outlines the problem statement, modeling approach, key results, and insights.
  - Discussion of model strengths, limitations, and implications for transportation planning or LBS data analysis.
- **Optional Extensions (time-permitting):**
  - Incorporation of exogenous features (e.g., weather, holidays).
  - Extension to multivariate or multi-zone modeling with spatiotemporal attention.
  - Sensitivity analysis on data sparsity or temporal coverage.

6 Timeline

- **Week 1-2: Dataset Exploration and Baseline Modeling**
  - Load and clean the NYC Yellow Taxi Trip dataset.
  - Aggregate data into hourly zone-level demand.
  - Perform exploratory data analysis (EDA) to identify temporal patterns and spatial variation.
  - Implement baseline LSTM model using lag features and temporal embeddings.
  - Evaluate preliminary performance using MAE, RMSE, and visualizations.
- **Week 3-4: Transformer Modeling and Comparative Analysis**
  - Implement Transformer encoder model with fixed or learnable positional encodings.
  - Train and evaluate Transformer on the same input structure as LSTM.
  - Compare model performance across metrics and visual plots.
  - Refine temporal and spatial feature engineering based on error analysis.
  - (Optional) Experiment with spatial embeddings or attention over zones.
- **Week 5: Finalization and Reporting**
  - Tune hyperparameters and finalize model configurations.
  - Generate final evaluation plots and error breakdowns.
  - Prepare and polish the GitHub repository (code, documentation, dependencies).
  - Write and submit the 2-page summary report.

References

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